KENYA MEDICAL TRAINING COLLEGE

FACULTY OF DIAGNOSTIC SCIENCES

DEPARTMENT OF MEDICAL LABORATORY SCIENCES

DIPLOMA IN MEDICAL LABORATORY SCIENCES

YEAR I, SEMESTER I EXAMINATION (1.1)

PAPER: BIOCHEMISTRY

DATE: 10/02/2025 **TIME: 2 HOURS 9.00 – 11 AM**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple choice questions)
   - Section 2 (8 short answer questions)
   - Section 3 (2 long answer questions)
2. Attempt ALL Questions
3. Write the COLLEGE NUMBER on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Any examination malpractices will be handled as per the College Examination Policy
6. Ensure you sign the examination register provided

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SECTION 1: MULTIPLE CHOICE QUESTIONS (20MKS)

1. The major pathway for glucose metabolism is:
   A. Kreb's cycle
   B. Oxidative phosphorylation
   C. Hexose monophosphate shunt
   D. Glycolysis
2. The end products of glycolysis are:
   A. ATP only
   B. Pyruvate only
   C. ATP and Pyruvate
   D. ATP, pyruvate and NADH
3. The fat - soluble vitamins are:
   A. A, D, E and K
   B. Vit B1
   C. Vit B2
   D. Vitamin C
4. The primary input for Krebs cycle is:
   A. Acetyl CoA
   B. Pyruvate
   C. NADH
   D. FADH
5. The basic constituent of proteins is:
   A. Fatty acids
   B. Peptides
   C. Amino acids
   D. Monosaccharides
6. The bond formed by the joining of two or more amino acids is known as?
   A. Covalent bond
   B. Glycosidic bond
   C. Peptide bond
   D. Ionic bond
7. When protein molecules are put in a buffer solution of pH basic to their isoelectric point, they ionize and become:
   A. Positively charged
   B. Negatively charged
   C. Zwitterion
   D. Dielectric

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8. Phenylketonuria is a metabolic disorder resulting from deficiency of the enzyme:
   A. Hydroxylase
   B. Phenylalanine hydroxylase
   C. Tyrosine synthetase
   D. Peptidase
9. Which of the following is a conjugated protein?
   A. Lipoprotein
   B. Albumin
   C. Globulin
   D. Fibrinogen
10. The trace element needed for the synthesis of haemoglobin in the body is:
   A. Iodine
   B. Cobalt
   C. Zinc
   D. Iron
11. Which of the following biological molecules consists of amino acids?
   A. Carbohydrates
   B. Proteins
   C. Lipids
   D. Vitamins
12. The storage form of carbohydrates in body cells is:
   A. Glycogen
   B. Glucose
   C. Cellulose
   D. Starch
13. Identify among the following a monosaccharide attributed to diabetes mellitus:
   A. Fructose
   B. Galactose
   C. Xylose
   D. Glucose
14. Lactose is composed of which monosaccharides:
   A. Glucose and fructose
   B. Glucose and galactose
   C. Glucose and sucrose
   D. Glucose and glucose
15. The pentose sugar found abundant in the nuclei of cells is:
   A. Xylose
   B. Arabinose
   C. Ribose
   D. Fructose

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16. Which one of the following is a conjugated protein?
    A. Glycoprotein
    B. Fatty acid
    C. Cholesterol
    D. Hormone
17. The bond that joins monosaccharides together to form disaccharides is known as:
    A. Peptide bond
    B. Glycosidic bond
    C. Covalent bond
    D. Van der Waals forces
18. The molecular weight for glucose molecule ($C_6H_{12}O_6$) is:
    A. 100
    B. 180
    C. 80
    D. 120
19. Which of the following will reduce Benedict's reagent?
    A. Sucrose
    B. Glucose
    C. Starch
    D. Cellulose
20. The term that describes the degradation of glycogen to glucose is:
    A. Gluconeogenesis
    B. Glycogenesis
    C. Glycogenolysis
    D. Anabolism

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

21. Outline five (5) Classes of enzyme as per the IUB classification (5maks)
22. State five (5) applications of biochemistry in the field of medicine (5marks)
23. Outline five (5) biomolecules of medical importance (5marks)
24. Illustrate and label the general structure of amino acids (5marks)
25. Name five (5) properties of enzymes (5marks)
26. State five (5) functions of proteins in the body (5marks)
27. Enumerate the functions of cholesterol in the body (5marks)
28. Enumerate five (5) protein binding hormones (5marks)

SECTION 3: LONG ANSWER QUESTIONS ((40 MARKS)

29. Discuss the IUB classification biological catalysts (20mks)
30. With the help of graphical illustrations where applicable, discuss any Four (4) factors that may affect enzyme activity (20 marks)

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